

Table 2. Progressive model building reveals sex-specific genetic effects

Each row adds complexity to the previous model

Model Components	ΔR^2	Key Finding	Statistical Support
Selected Model (Model 3) Key Parameters			
Genetics only (HI, He)	0.019	No effect	$F_{3,300} = 1.94, p = 0.122$
+ Sex + Infection	0.044	Infection effect emerges	$F_{4,299} = 5.05, p < 0.001$
+ All two-way interactions	0.030	Sex $\times$ HI interaction¹	$F_{10,293} = 2.99, p = 0.001$
+ Three-way interactions	0.019	No improvement	$F_{15,288} = 2.42, p = 0.003$

¹ Infection effect: $\beta = 2.32\%$ (SE = 0.80, $p = 0.004$); Sex $\times$ HI: $\beta = 2.14$ (SE = 0.83, $p = 0.011$)

Interpretation: Males show increasing predicted weight loss with *M. m. musculus* ancestry; females do not